

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Networks (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Network. Explain the role of Internet Service Providers (ISPs) and Network Service Providers (NSPs). List the merits of client-server model over peer to peer model. 7
b) Draw the TCP/IP model and explain one of the protocols of each layer. 8
2. a) Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter. 8
b) What is flow control? Describe Stop-and-Wait, Go-Back-N, and Selective Repeat protocols with suitable diagrams. 7
3. a) Illustrate cyclic redundancy check method for error detection with an example. 8

OR

Explain the algorithm of CSMA/CD and draw the flowchart.

- b) An organization is planning to design 6 different branches for Marketing, IT, AI, Management, Accounting and Software. Each department consists of 28 computers with IP address 192.108.254.0. Explain the process of allocating the IP address and subnet for each of the departments using above IP. 7
4. a) What is classful and classless address? Differentiate between link state and distance vector routing protocol. 7

OR

Explain IPv4 header format. Differentiate between IPv4 and IPv6.

- b) What are the function of transport layer. Explain the concept of socket programming in the transport layer protocol. 8
5. a) Explain the process of token based leaky bucket algorithm to address congestion during packet propagation. 5

- b) What are the roles of Domain Name Systems? Explain the working principles of Dynamic Host Configuration Protocol (DHCP). 5
- c) Explain the importance of proxy server and firewall in local area network. 5
- 6. a) What is data encryption? Explain the role of symmetric and asymmetric key cryptography with a suitable example. 8
- b) What do you mean by network Management? Explain SNMP protocol. 7
- 7. Write short notes on: (**Any two**) 2×5
 - a) TCP congestion control
 - b) NAT
 - c) Switch versus router